

Sec 7.1/7.2 systems of 1st order ODEs

Ex x, y functions of t ($x' = \frac{dx}{dt}$, $y' = \frac{dy}{dt}$)

System of 1st order ODEs $\begin{cases} x' = y \\ y' = -x \end{cases}$ see that the functions and their derivatives (x, x', y, y') are "intertwined".

Often call this a system of "coupled" ODEs.

An "uncoupled" system example $\begin{cases} x' = 2x + 3 \\ y' = \frac{1}{y} \end{cases}$. "Coupled" are generally harder to solve.

General system of 1st order eqs (only two vars in this example) is $\begin{cases} x_1' = f(x_1, x_2, t) \\ x_2' = g(x_1, x_2, t) \end{cases}$

The system $\begin{cases} x_1' = x_2 \\ x_2' = -x_1 \end{cases}$ is in "scalar form"

We often like to work with systems in "vector form":

$$\begin{cases} x_1' = x_2 \\ x_2' = -x_1 \end{cases} \rightarrow \text{same as } \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0x_1 + 1x_2 \\ -1x_1 + 0x_2 \end{bmatrix}$$

"vector form" $\rightarrow \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Can also write $\underline{X}' = \underline{A} \underline{X}$

or $\underline{X}' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \underline{X}$

Ex: Write system $\begin{cases} x' = 2x - 3y + z + t \\ y' = y + 7z - t \\ z' = x - 4y + 8z + t^3 \end{cases}$
in "vector form".

★ keep "t-terms" as a separate vector.

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 2x - 3y + 1z \\ 0x + 1y + 7z \\ 1x - 4y + 8z \end{bmatrix} + \begin{bmatrix} t \\ -t \\ t^3 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}}_{\underline{X}'} = \underbrace{\begin{bmatrix} 2 & -3 & 1 \\ 0 & 1 & 7 \\ 1 & -4 & 8 \end{bmatrix}}_P \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\underline{X}} + \underbrace{\begin{bmatrix} t \\ -t \\ t^3 \end{bmatrix}}_{\underline{f}(t)} \quad (\text{vector form})$$

So general "vector form" is

$$\underline{X}' = P\underline{X} + \underline{f}(t)$$

Thus, we see that a system of $n \times 1^{\text{st}}$ order ODEs
corresponds to $\uparrow$ $(n \times 1)$ -vector ODE"
a single

• General 1^{st} order "vector ODE": $\underline{X}' = \underbrace{P}_{(n \times n)} \underline{X} + \underbrace{\underline{f}(t)}_{(n \times 1)}$

• If $\underline{f}(t) = \underline{0}$, or $\underline{f}(t)$ is missing, then eq. $\underline{X}' = P\underline{X}$ is said to be "homogeneous". If not, $\underline{X}' = P\underline{X} + \underline{f}(t)$ nonhomog.

• The matrix P can vary with time (see example below). If it is constant, often we write "A" instead of "P".

Higher order linear ODEs (like $y'' - y' - 2y = 0$) can be "converted" into a system of 1st order ODEs.

In General: an n -th order ODE $\longleftrightarrow n \times$ 1st order ODEs
(n variables)

We make this conversion by (essentially) make each derivative a new variable

Ex Convert $x'' + 3x' - 6x = t^2$ to system of 1st order ODEs.

Goal: 1x 2nd order $\longrightarrow$ 2x 1st orders eqs $\begin{cases} x_1' = \dots \\ x_2' = \dots \end{cases}$

Set $\begin{cases} x_1 = x \\ x_2 = x' \\ x_3 = x'' \end{cases} \longrightarrow \begin{cases} x_1' = x' = x_2 \\ x_2' = x'' = ? \end{cases}$ (want to relate x'' to x, x', t)
 ~~$x_3 = x''$~~ $\rightarrow$ don't need x_3

To connect x'' to lower order derivatives, use ODE given:

$$x'' + 3x' - 6x = t^2 \Rightarrow x'' = 6x - 3x' + t^2$$
$$x'' = \underline{6x_1 - 3x_2 + t^2}$$

So total system is $\begin{cases} x_1' = x_2 \\ x_2' = 6x_1 - 3x_2 + t^2 \end{cases}$ (scalar form)

vector form: $\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}$

Ex: Convert $x^{(4)} + 3x^{(3)} + 2x'' - x' - 7x = 0$
into a system of 1st order ODEs.

4th order $\rightarrow$ 4x 1st orders, $\begin{cases} x_1' = \\ x_2' = \\ x_3' = \\ x_4' = \end{cases}$

SET $\begin{cases} x_1 = x \\ x_2 = x' \\ x_3 = x'' \\ x_4 = x^{(3)} \\ \cancel{x_5 = x^{(4)}} \end{cases} \Rightarrow \begin{cases} x_1' = x' = x_2 \\ x_2' = x'' = x_3 \\ x_3' = x^{(3)} = x_4 \\ x_4' = x^{(4)} = 7x_1 + x_2 - 2x_3 - 3x_4 \end{cases}$ } scalar form

don't need x_5

use $x^{(4)} + 3x^{(3)} + 2x'' - x' - 7x = 0$
 $\Rightarrow x^{(4)} = 7x + x' - 2x'' - 3x^{(3)}$
 $x^{(4)} = 7x_1 + x_2 - 2x_3 - 3x_4$

vector form: $\begin{bmatrix} x_1' \\ x_2' \\ x_3' \\ x_4' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 7 & 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ (homog. $\underline{X}' = \underline{A}\underline{X}$)

"multiple higher orders": Ex: Convert system of 2nd order ODEs $\begin{cases} x'' - 5x' + 4x - 3y = 0 \\ y'' - 7y' + 3x + 2y = 0 \end{cases}$ into system of 1st order ODEs.
(2x 2nd order $\rightarrow$ 4x 1st orders)

Set $\begin{cases} z_1 = x \\ z_2 = x' \\ z_3 = y \\ z_4 = y' \end{cases} \Rightarrow \begin{cases} z_1' = x' = z_2 \\ z_2' = x'' = -4x + 5x' + 3y + 0y' = -4z_1 + 5z_2 + 3z_3 \\ z_3' = y' = z_4 \\ z_4' = y'' = -3x + 0x' - 2y + 7y' = -3z_1 - 2z_3 + 7z_4 \end{cases}$

In total: $\begin{cases} z_1' = z_2 \\ z_2' = -4z_1 + 5z_2 + 3z_3 + 0z_4 \\ z_3' = z_4 \\ z_4' = -3z_1 - 2z_3 + 7z_4 \end{cases}$ (scalar form)

$\begin{bmatrix} z_1' \\ z_2' \\ z_3' \\ z_4' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -4 & 5 & 3 & 0 \\ 0 & 0 & 0 & 1 \\ -3 & 0 & -2 & 7 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \end{bmatrix}$ (vector form)

Ex Convert the ODE $t^2 x'' + tx' + (t^2 - k^2)x = 0$ (Bessel Eq) (k const) into a system of 1st order ODEs.

Rewrite equation as $x'' = \frac{k^2 - t^2}{t^2} x - \frac{t}{t^2} x' = \frac{k^2 - t^2}{t^2} x - \frac{1}{t} x'$

SET $\begin{cases} x_1 = x \\ x_2 = x' \end{cases} \Rightarrow \begin{cases} x_1' = x_2 \\ x_2' = x'' = \frac{k^2 - t^2}{t^2} x_1 - \frac{1}{t} x_2 \end{cases}$ scalar form

vector form: $\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{k^2 - t^2}{t^2} & -\frac{1}{t} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ ($\underline{X}' = P(t)\underline{X}$, homog.)

Ex: Find a particular solution for $\underline{X}' = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \underline{X}$ which has $\underline{X}(0) = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ (recall eigenvalues/vectors: $\lambda_1 = 0, \underline{v}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$
 $\lambda_2 = 5, \underline{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$)

General solution

$$\underline{X}_1 = e^{\lambda_1 t} \underline{v}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\underline{X}_2 = e^{\lambda_2 t} \underline{v}_2 = e^{5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\underline{X} = c_1 \underline{X}_1 + c_2 \underline{X}_2 = \begin{bmatrix} -2c_1 + c_2 e^{5t} \\ c_1 + 2c_2 e^{5t} \end{bmatrix} \begin{matrix} \leftarrow "x_1(t)" \\ \leftarrow "x_2(t)" \end{matrix}$$

$$\underline{X}(0) = \begin{bmatrix} -2c_1 + c_2 \\ c_1 + 2c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

... $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -7/5 \\ 1/5 \end{bmatrix}$, so particular solution is

$$\underline{X} = -\frac{7}{5} \underline{X}_1 + \frac{1}{5} \underline{X}_2 = \frac{1}{5} \begin{bmatrix} 14 + e^{5t} \\ -7 + 2e^{5t} \end{bmatrix}$$

For this example with $\underline{X}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $\underline{X}_2 = e^{5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, we need to be sure that $\underline{X}_1, \underline{X}_2$ are linearly indep in order that the general solution actually is $c_1 \underline{X}_1 + c_2 \underline{X}_2$.

• In general, system of n -1st order eqs (aka $(n \times 1)$ -vector ODE) requires n linearly indep solutions $\underline{X}_1, \dots, \underline{X}_n$ $\downarrow$ on " n -th order eq"

to make general solution $\underline{X} = c_1 \underline{X}_1 + \dots + c_n \underline{X}_n$.

So how do we check linear indep?

★ $\underline{X}_1, \underline{X}_2, \dots, \underline{X}_n$ are linearly indep if lin combination

$c_1 \underline{X}_1 + \dots + c_n \underline{X}_n = \underline{0}$ is possible only with the trivial solution $(c_1, \dots, c_n) = (0, \dots, 0)$. Otherwise they are lin dependent.

$$\underline{X}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad \underline{X}_2 = e^{5t} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix}$$

$$\begin{aligned} c_1 \underline{X}_1 + c_2 \underline{X}_2 &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow c_1 \begin{bmatrix} -2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} e^{5t} \\ 2e^{5t} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \\ &\rightarrow \begin{bmatrix} -2 & e^{5t} \\ 1 & 2e^{5t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

$$\underline{A} \underline{x} = \underline{0}$$

Equation $\underline{A} \underline{x} = \underline{0}$ has only trivial solution $\underline{x} = \underline{0}$ if and only if $\underline{A}^{-1}$ defined, which happens if and only if $\det \underline{A} \neq 0$

$\det \left(\begin{bmatrix} -2 & e^{5t} \\ 1 & 2e^{5t} \end{bmatrix} \right) = -4e^{5t} - e^{5t} = -5e^{5t} \neq 0, \dots$, so, only solution $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$ is trivial solution $\begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \underline{X}_1, \underline{X}_2$ are lin indep.

"DEF" determinant $\det \left(\begin{bmatrix} -2 & e^{5t} \\ 1 & 2e^{5t} \end{bmatrix} \right) = \det \left(\begin{bmatrix} \underline{X}_1 & \underline{X}_2 \\ | & | \end{bmatrix} \right) = W(\underline{X}_1, \underline{X}_2)$
"Wronskian"

DEF: If $\underline{X}_1, \dots, \underline{X}_n$ are $(n \times 1)$ -vector functions, then Wronskian

$$W(\underline{X}_1, \dots, \underline{X}_n) \stackrel{\text{DEF}}{=} \det \left(\begin{bmatrix} | & | & | \\ \underline{X}_1 & \underline{X}_2 & \dots & \underline{X}_n \\ | & | & | \end{bmatrix} \right)$$

★ Note: When we did Wronskian for functions $y_1, \dots, y_n$ we made n -rows total using derivatives $y_1', y_2', \dots$
 $y_1'', \dots$

- When dealing with vector functions $\underline{X}_1, \dots, \underline{X}_n$, matrix $\begin{bmatrix} \uparrow \underline{X}_1 & \dots & \uparrow \underline{X}_n \\ \downarrow & & \downarrow \end{bmatrix}$ is already $(n \times n)$; no derivatives needed.
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Thm: If $\underline{X}_1, \dots, \underline{X}_n$ are $(n \times 1)$ -vector functions, then they are linearly independent $\iff W(\underline{X}_1, \dots, \underline{X}_n) \neq 0$.